MILITARY SPECIFICATION SHEET

WAVEGUIDE ASSEMBLIES, EXTRA-FLEXIBLE, TWISTABLE, CLASS 3

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the waveguide assemblies described herein shall consist of this document and the latest issue of Specification MIL-W-287

REQUIREMENTS: See table I.

Part number: See table I.

Extent of qualification: Qualification obtained for one assembly shall constitute qualification for other assemblies as specified in table II.

Cross reference information See table III

Substitution data See table IV

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TABLE I Extra-flexible.

Group number	1/ Part number M287/5-	TE_{10} mode frequency range (GHz)	Minimum centerline bending radius (inch)		Repeated twist (degrees/ft)		Insertion loss (dB/ft)	Axial twist (degrees/ft)	Pressurization lb_f/in^2 gage	2/ VSWR (max) at			
			E plane	H plane	400 cycles	$1x10^5$ cycles				12"	24"	36"	>36"
I	01-XXXX 02-XXXX 60-XXXX	1 12- 1 70	13 02	27 07	27	6	0 025	54	2	1 05	1 06	1 07	1 10
	03-XXXX 04-XXXX 61-XXXX	1.70- 2 60	9 51	14 00	30	9	0 035	60	20	1 06	1 07	1 08	1 10
	05-XXXX 06-XXXX 62-XXXX	2 20- 3 30	6 00	8 00	35 0	10 5	0 05	70	30	1 06	1 07	1 08	1 10
	07-XXXX 08-XXXX 09-XXXX 10-XXXX 11-XXXX 12-XXXX 13-XXXX 14-XXXX 15-XXXX 63-XXXX	2.60- 3 95	4 50	6 38	42 0	12 6	0 06	84	30	1 08	1 09	1 10	1 12
II	16-XXXX 17-XXXX 18-XXXX 19-XXXX 20-XXXX 21-XXXX 22-XXXX 23-XXXX 64-XXXX	3 95- 5 85	3 00	4 75	67 50	20 25	0 07	135	45	1 08	1 09	1 10	1 12
	24-XXXX 25-XXXX 26-XXXX 27-XXXX 28-XXXX 29-XXXX 30-XXXX 31-XXXX 65-XXXX	5 85- 8 20	2 50	4 75	70	21	0 08	140	45	1 08	1 09	1 10	1 12
	32-XXXX 33-XXXX 34-XXXX 35-XXXX 36-XXXX 37-XXXX 38-XXXX 39-XXXX 66-XXXX	7 05- 10 0	2 00	4 00	90	27	0 08	180	45	1 12	1 14	1 15	1 18

See footnotes at end of table

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twistable waveguide assemblies

Flexure (degrees/ft) 1x10^5 cycles		Jacket material	Retention force (lb_f)	Oper temp range -55 to	Mates with standard rigid waveguide M85/1-	Flanges		
E plane	H plane					Flange 1 M3922/( )	Flange 2 M3922/( )	Material 3/
7 2	3 6	Neoprene	N/A	+100°C	017	58-007 (contact)	58-007 (contact)	CU
7 2	3 6				018	58-008 (Contact)	58-008 (contact)	AL
4/	4/				017	57-007 (contact)	57-007 (contact)	CU
8 6	4 3	Neoprene	N/A	+100°C	031	58-009 (contact)	58-009 (contact)	CU
8 6	4 3				029	58-010 (contact)	58-010 (contact)	AL
4/	4/				031	58-009 (contact)	58-009 (contact)	CU
10 50	5 28	Neoprene	N/A	+100°C	037	58-011 (contact)	58-011 (contact)	CU
10 50	5 28				035	58-012 (contact)	58-012 (contact)	AL
4/	4/				037	58-011 (contact)	58-011 (contact)	CU
11 5	7 5	Neoprene	N/A	+100°C	043	64-001 (contact)	64-001 (contact)	CU
					041	64-002 (contact)	64-002 (contact)	AL
					043	56-001 (cover)	56-001 (cover)	CU
					041	56-002 (cover)	56-002 (cover)	AL
					043	61-002 (choke)	56-001 (cover)	CU
					041	61-001 (choke)	56-002 (cover)	AL
					043	61-002 (choke)	61-002 (choke)	CU
					041	61-001 (choke)	61-001 (choke)	AL
4/	4/	Plastic		+65°C	041	52-032 (contact)	52-032 (contact)	AL
		Neoprene		+100°C	043	58-011 (contact)	58-011 (contact)	CU
19 0	9 5	Neoprene	N/A	+100°C	055	63-001 (contact)	63-001 (contact)	CU
					053	63-005 (contact)	63-005 (contact)	AL
					055	57-002 (cover)	57-002 (cover)	CU
					053	57-001 (cover)	57-001 (cover)	AL
					055	62-002 (choke)	57-002 (cover)	CU
					053	62-001 (choke)	57-001 (cover)	AL
					055	62-002 (choke)	62-002 (choke)	CU
					053	62-001 (choke)	62-001 (choke)	AL
4/	4/				055	63-001 (contact)	63-001 (contact)	CU
26 50	13 25	Neoprene	N/A	+100°C	067	63-002 (contact)	63-002 (contact)	CU
					065	63-006 (contact)	63-006 (contact)	AL
					067	55-001 (cover)	55-001 (cover)	CU
					065	55-002 (cover)	55-002 (cover)	AL
					067	60-001 (choke)	55-001 (cover)	CU
					065	60-002 (choke)	55-002 (cover)	AL
					067	60-001 (choke)	60-001 (choke)	CU
					065	60-002 (choke)	60-002 (choke)	AL
4/	4/				067	63-002 (contact)	63-002 (contact)	CU
28.8	14 4	Neoprene	N/A	+100°C	073	63-003 (contact)	63-003 (contact)	CU
					071	63-007 (contact)	63-007 (contact)	AL
					073	53-002 (cover)	53-002 (cover)	CU
					071	53-004 (cover)	53-004 (cover)	AL
					073	59-007 (choke)	53-002 (cover)	CU
					071	59-009 (choke)	53-004 (cover)	AL
					073	59-007 (choke)	59-007 (choke)	CU
					071	59-009 (choke)	59-009 (choke)	AL
4/	4/				073	63-003 (contact)	63-003 (contact)	CU

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TABLE I Extra-flexible, twistable

Group number	1/ Part number M287/5-	TE_{10} mode frequency range (GHz)	Minimum centerline bending radius (inch)		Repeated twist (degrees/ft)		Insertion loss (dB/ft)	Axial twist (degrees/ft)	Pressurization lb_f/in^2 gage	2/ VSWR (max) at			
			E plane	H plane	400 cycles	$1x10^5$ cycles				12"	24"	36"	>36"
II	40-XXXX 41-XXXX 42-XXXX 43-XXXX 44-XXXX 45-XXXX 46-XXXX 47-XXXX 67-XXXX	8 2- 12.4	1 38	2 25	120	36	0 13	240	60	1 14	1 15	1 16	1 19
III	48-XXXX 49-XXXX 50-XXXX 51-XXXX 52-XXXX 53-XXXX 68-XXXX	12 4- 18 0	1 25	1 75	120	36	0 30	250	60	1 13	1 15	1 19	1 24
	54-XXXX 55-XXXX 56-XXXX 57-XXXX 58-XXXX 59-XXXX 69-XXXX	18 0 26 5	1 12	1 25	132	30	0 35	264	60	1 14	1 16	1 20	1 25

1/ The complete part number shall be the number shown and a four-digit number indicating the nominal relaxed length in inches
2/ The VSWR value for any length between the lengths specified shall be the next higher value (e g for a 15 inch length of M287/5-01, the VSWR would be 1 06 1)
3/ CU = copper alloy, AL = aluminum alloy
4/ The assemblies are constructed from beryllium copper and the flexure cycle is 10^6 rather than 10^5 All other characteristics are as listed

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waveguide assemblies - Continued

Flexure (degrees/ft) $1x10^5$ cycles		Jacket material	Retention force (lb_f)	Oper temp range -55 to	Mates with standard rigid waveguide M85/1-	Flanges		
E plane	H plane					Flange 1 M3922/( )	Flange 2 M3922/( )	Material 3/
34 25	17 12	Neoprene	N/A	+100°C	079	63-004 (contact)	63-004 (contact)	CU
					077	63-008 (contact)	63-008 (contact)	AL
					079	53-001 (cover)	53-001 (cover)	CU
					077	53-003 (cover)	53-003 (cover)	AL
					079	59-006 (choke)	53-001 (cover)	CU
					077	59-008 (choke)	53-003 (cover)	AL
					079	59-006 (choke)	59-006 (choke)	CU
					077	59-008 (choke)	59-008 (choke)	AL
4/	4/				079	63-004 (contact)	63-004 (contact)	CU
53 [illegible]	26 5	Neoprene	N/A	+100 C	089	53-005 (cover)	53-005 (cover)	CU
					090	53-006 (cover)	53-006 (cover)	AL
					089	59-001 (choke)	53-005 (cover)	CU
					090	59-002 (choke)	53-006 (cover)	AL
					089	59-001 (choke)	59 001 (choke)	CU
					090	59-002 (choke)	59-002 (choke)	AL
4	4/				089	53 005 (cover)	53-005 (cover)	CU
[illegible] 5	34 5	Neoprene	N/A	+100°C	102	54-001 (cover)	54-001 (cover)	CU
					103	54-002 (cover)	54-002 (cover)	AL
					102	59-003 (choke)	54-001 (cover)	CU
					103	59-004 (choke)	54-002 (cover)	AL
					102	59-003 (choke)	59-003 (choke)	CU
					103	59-004 (choke)	59-004 (choke)	AL
4	4/				102	54 001 (cover)	54-001 (cover)	CU

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TABLE II. Grouping for qualification.

Group number	Qualifying dash number	Dash numbers of assemblies qualified
I	04	01, 02, 03, and 05 thru 15
I	61	60, 62, and 63
II	25	16 thru 24 and 26 thru 47
II	65	64, 66, and 67
III	48	49 thru 53
III	55	54, 56 thru 59
III	69	68

TABLE III. Cross reference (type designation to part number).

Type designation	Part number M287/5-( )	Type designation	Part number M287/5-( )
FG650CEEB	01	FG042CCCB	56
FG430CEEB	03	FG284CCCA	12
FG340CEEB	05	FG187CCCA	21
FG284CEEB	07	FG137CCCA	29
FG187CEEB	16	FG112CCCA	37
FG137CEEB	24	FG090CCCA	45
FG112CEEB	32	FG062CCCA	51
FG090CEEB	40	FG042CCCA	57
FG650CEEA	02	FG284CDDB	09
FG430CEEA	04	FG187CDDB	18
FG340CEEA	06	FG137CDDB	26
FG284CEEA	08	FG112CDDB	34
FG187CEEA	17	FG090CDDB	42
FG137CEEA	25	FG062CDDB	48
FG112CEEA	33	FG042CDDB	54
FG090CEEA	41	FG284CDDA	10
FG284CCCB	11	FG187CDDA	19
FG187CCCB	20	FG137CDDA	27
FG137CCCB	28	FG112CDDA	35
FG112CCCB	36	FG090CDDA	43
FG090CCCB	44	FG062CDDA	49
FG062CCCB	50	FG042CDDA	55

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TABLE IV Substitution data 1/

Dash number	Manufacturer's code (FSCM)	Manufacturer's part number
01	80006	AT-2B-650
01	87864	TFT-1020000
02	80006	AT-6B-650
02	87864	TFT-102001
03	80006	AT-2B-430
03	87864	TFT-94000
04	80006	AT-6B-430
04	87864	TFT-94001
05	80006	AT-2B-340
05	87864	TFT-89000
06	80006	AT-6B-340
06	87864	TFT-89001
07	80006	AT-12B-284
07	87864	TFT-86055
08	80006	AT-16B-284
08	87864	TFT-86056
09	80006	AT-2B-284
09	87864	TFT-86059
10	80006	AT-6B-284
10	87864	TFT-86060
11	80006	AT-3B-284
11	87864	TFT-86057
12	80006	AT-7B-284
12	87864	TFT-786058
13	80006	AT-4B-284
14	80006	AT-8B-284
16	80006	AT-12B-187
16	87864	TFT-78044
17	80006	AT-16B-187
17	87864	TFT-78045
18	80006	AT-2B-187
18	87864	TFT-78048
19	80006	AT-6B-187
19	87864	TFT-78049
20	80006	AT-3B-187
20	87864	TFT-78046
21	80006	AT-7B-187
21	87864	TFT-78047
22	80006	AT-4B-187
23	80006	AT-8B-187
24	80006	AT-12B-137
24	87864	TFT-72081
25	80006	AT-16B-137
25	87864	TFT-72082
26	80006	AT-2B-137
26	87864	TFT-72085
27	80006	AT-6B-137
27	87864	TFT-72086
28	80006	AT-3B-137
28	87864	TFT-72083
29	80006	AT-7B-137
29	87864	TFT-72084
30	80006	AT-4B-137
31	80006	AT-8B-137
32	80006	AT-12B-112
32	87864	TFT-68065
33	80006	AT-16B-112
33	87864	TFT-68066
34	80006	AT-2B-112
34	87864	TFT-68011

See footnote at end of table

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TABLE IV. Substitution data 1/ -Continued

Dash number	Manufacturer's code (FSCM)	Manufacturer's part number
35	80006	AT-6B-112
35	87864	TFT-68068
36	80006	AT-3B-112
36	87864	TFT-68014
37	80006	AT-7B-112
37	87864	TFT-68067
38	80006	AT-4B-112
39	80006	AT-8B-112
40	80006	AT-12B-090
40	87864	TFT-64139
41	80006	AT-16B-090
41	87864	TFT-64140
42	80006	AT-2B-090
42	87864	TFT-64141
43	80006	AT-6B-090
43	87864	TFT-64104
44	80006	AT-3B-090
44	87864	TFT-64094
45	80006	AT-7B-090
45	87864	TFT-64061
48	80006	AT-2B-062
48	87864	TFT-57036
49	80006	AT-6B-062
49	87864	TFT-57041
50	80006	AT-3B-062
50	87864	TFT-57050
51	80006	AT-7B-062
51	87864	TFT-57040
52	80006	AT-4B-062
53	80006	AT-8B-062

1/ Assemblies covered by this specification sheet are substitutable for the manufacturer's part numbers shown. This information in no way implies that the manufacturer's part is substitutable for the military part

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - AV, MI
Navy - OS
Air Force - 17, 99
DLA - ES

User activities
Army - AR
Navy - MC, CG, AS
Air Force - 11, 19

Preparing activity
Navy - EC

Agent
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